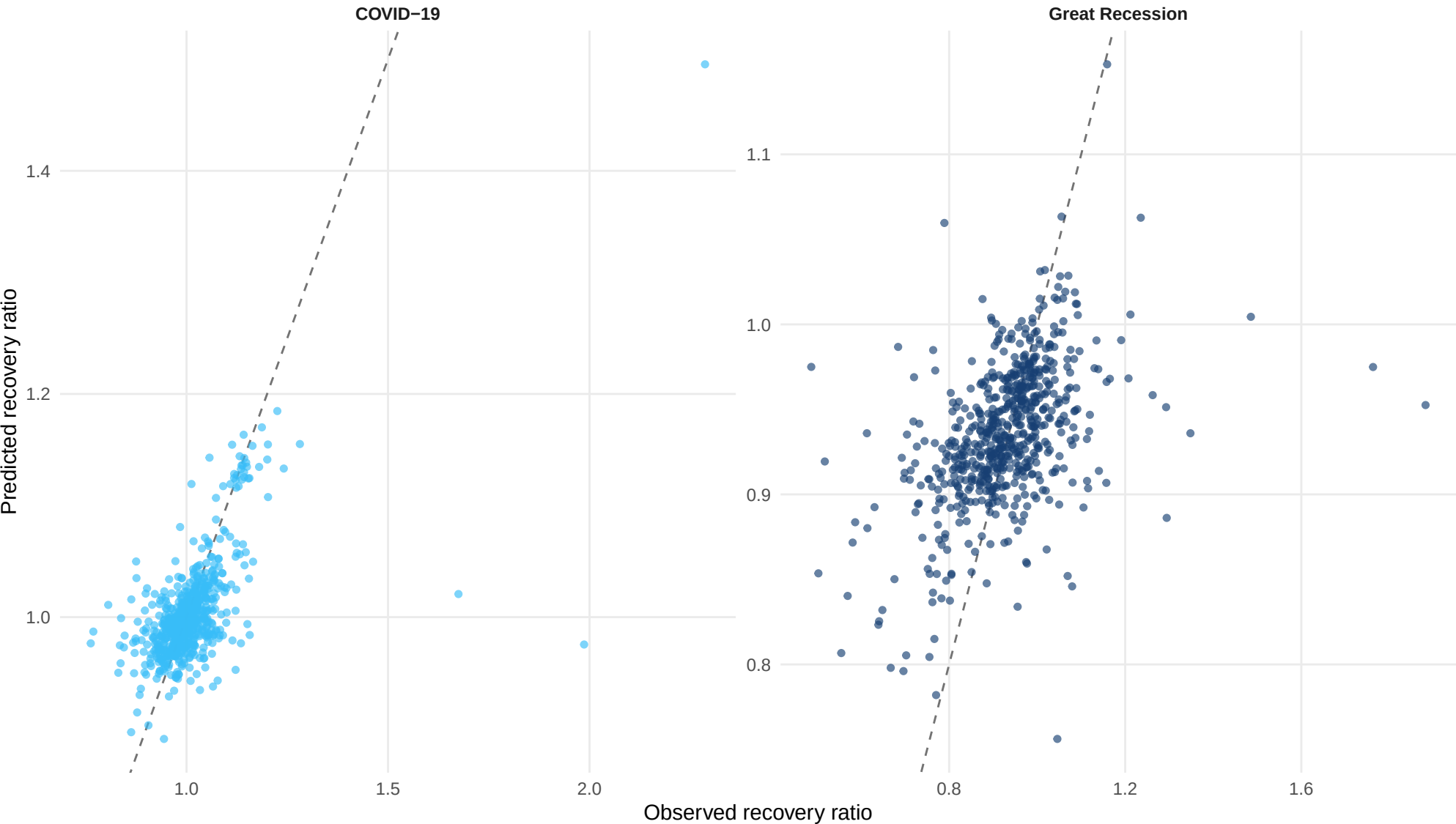


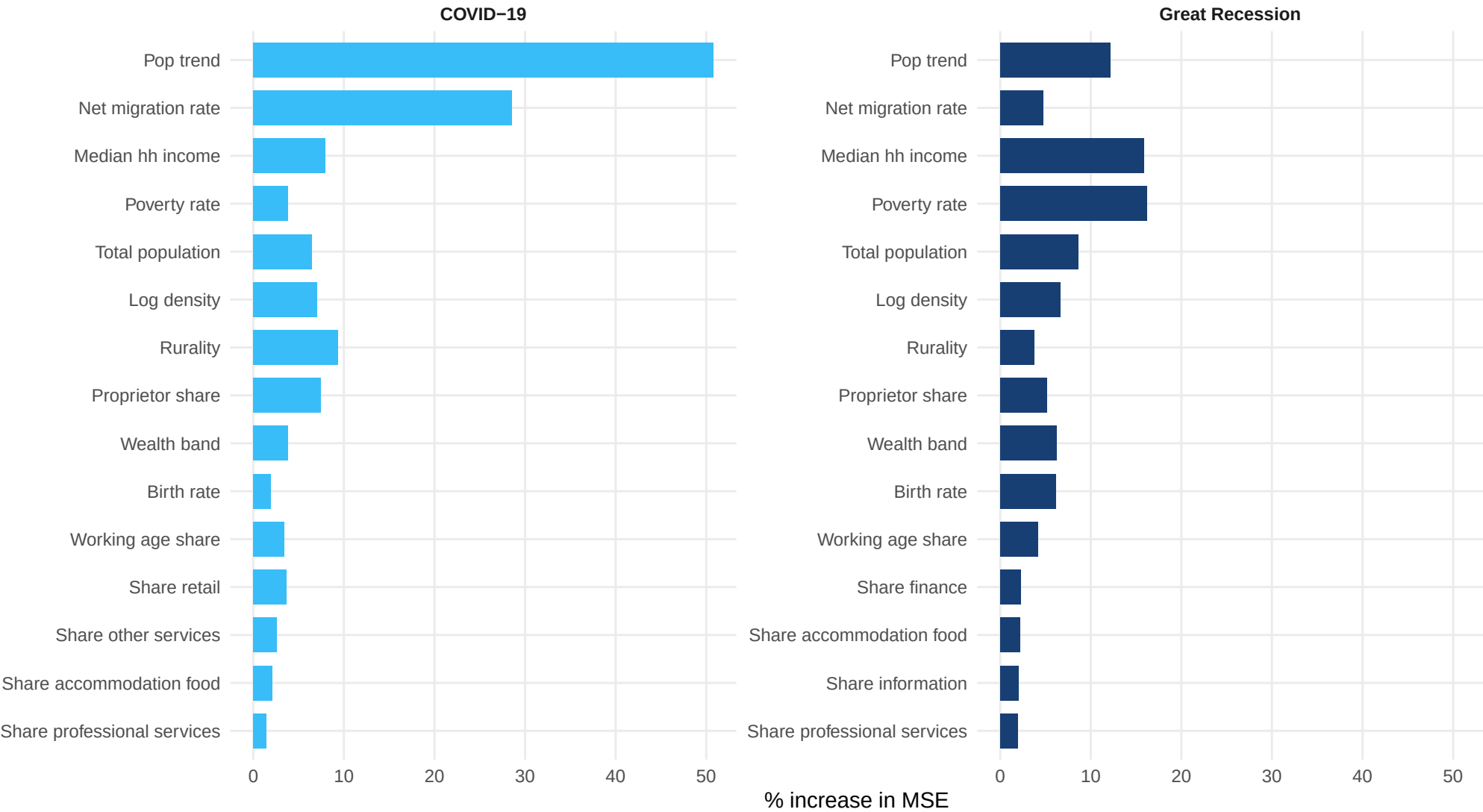
County random forest: observed vs predicted

20% holdout set



County variable importance

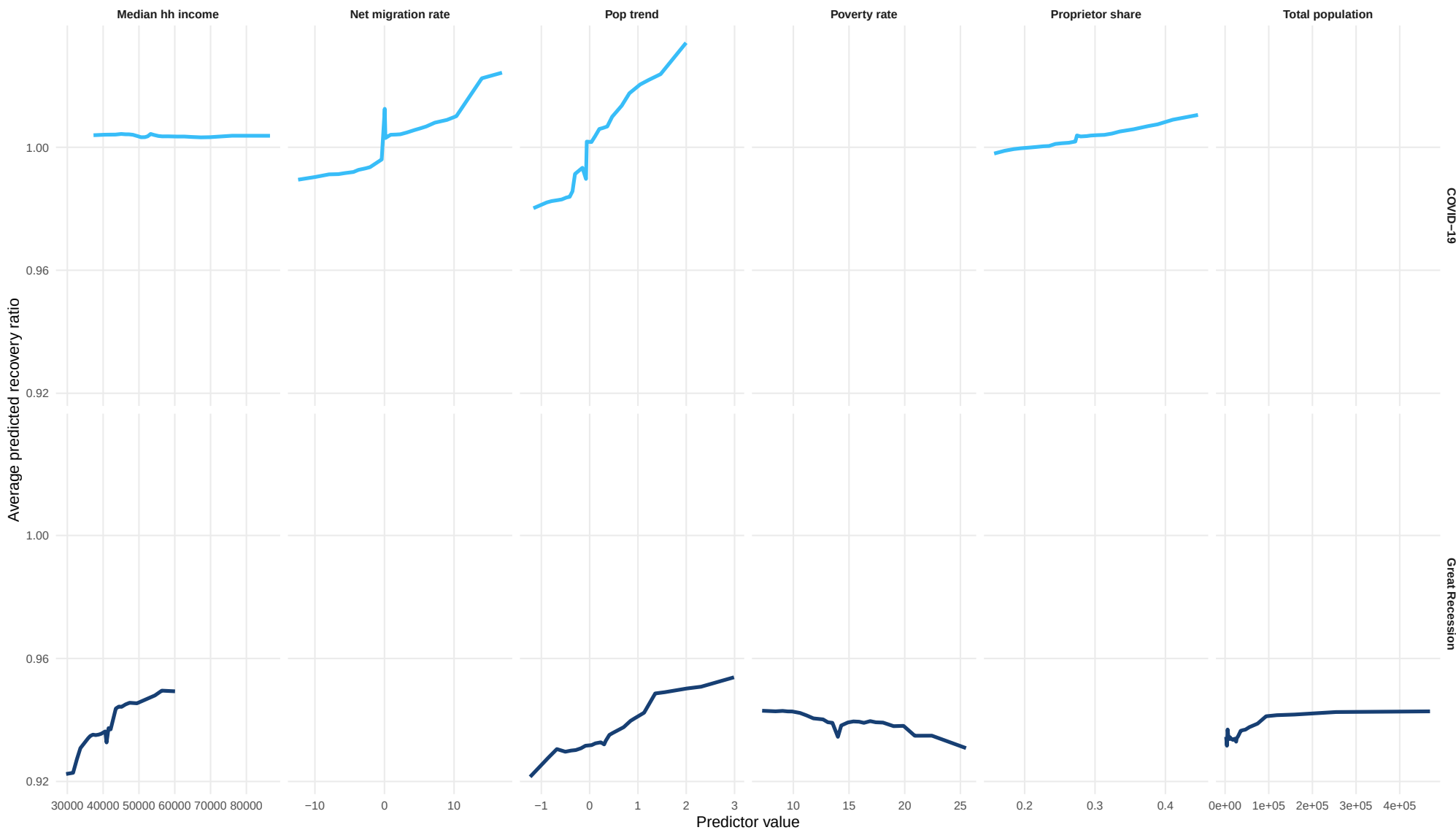
% increase in MSE when predictor is shuffled



Importance indicates predictive reliance, not causality.

County response profiles

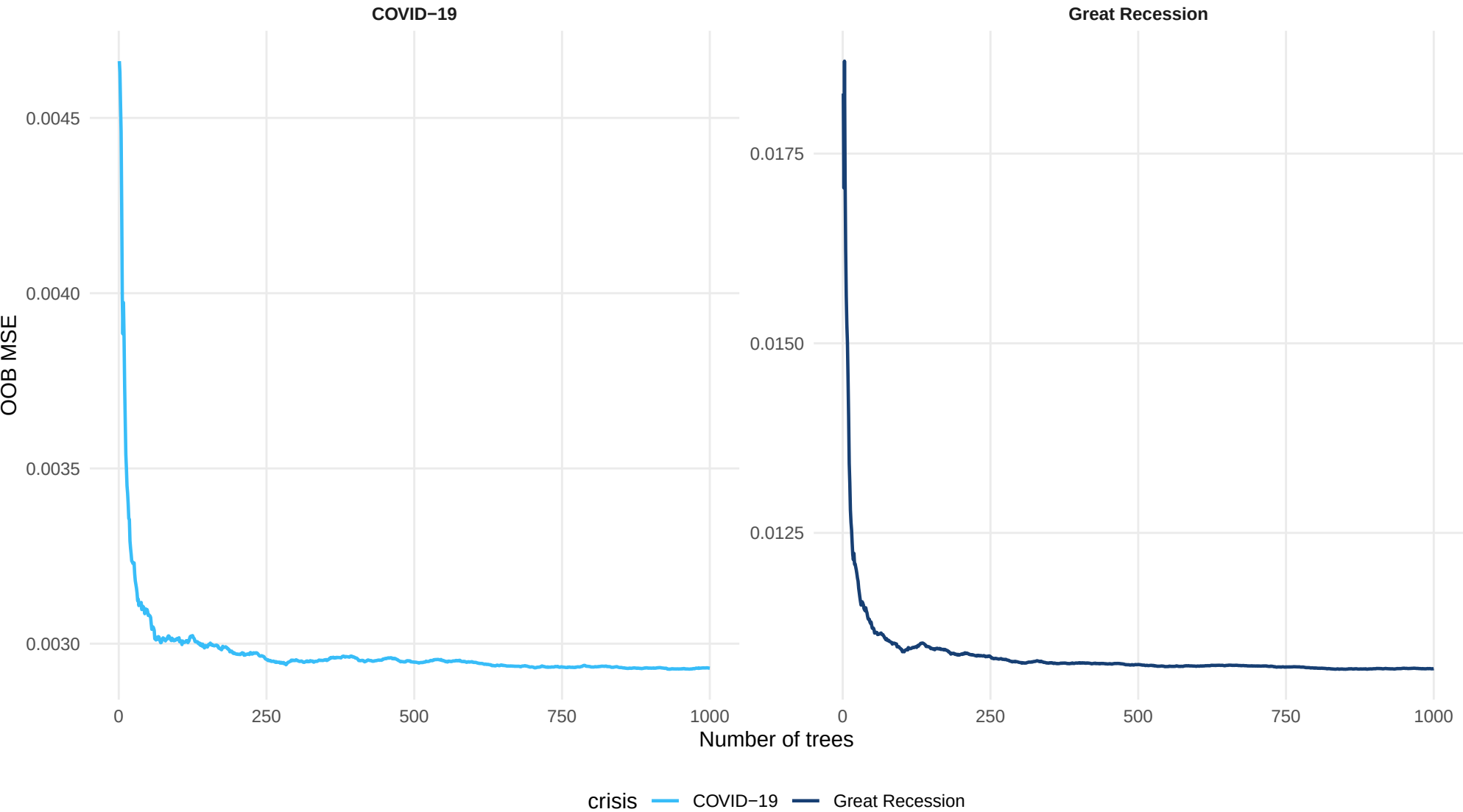
Marginal prediction curves for top numeric predictors



Curves describe model behaviour, not causation.

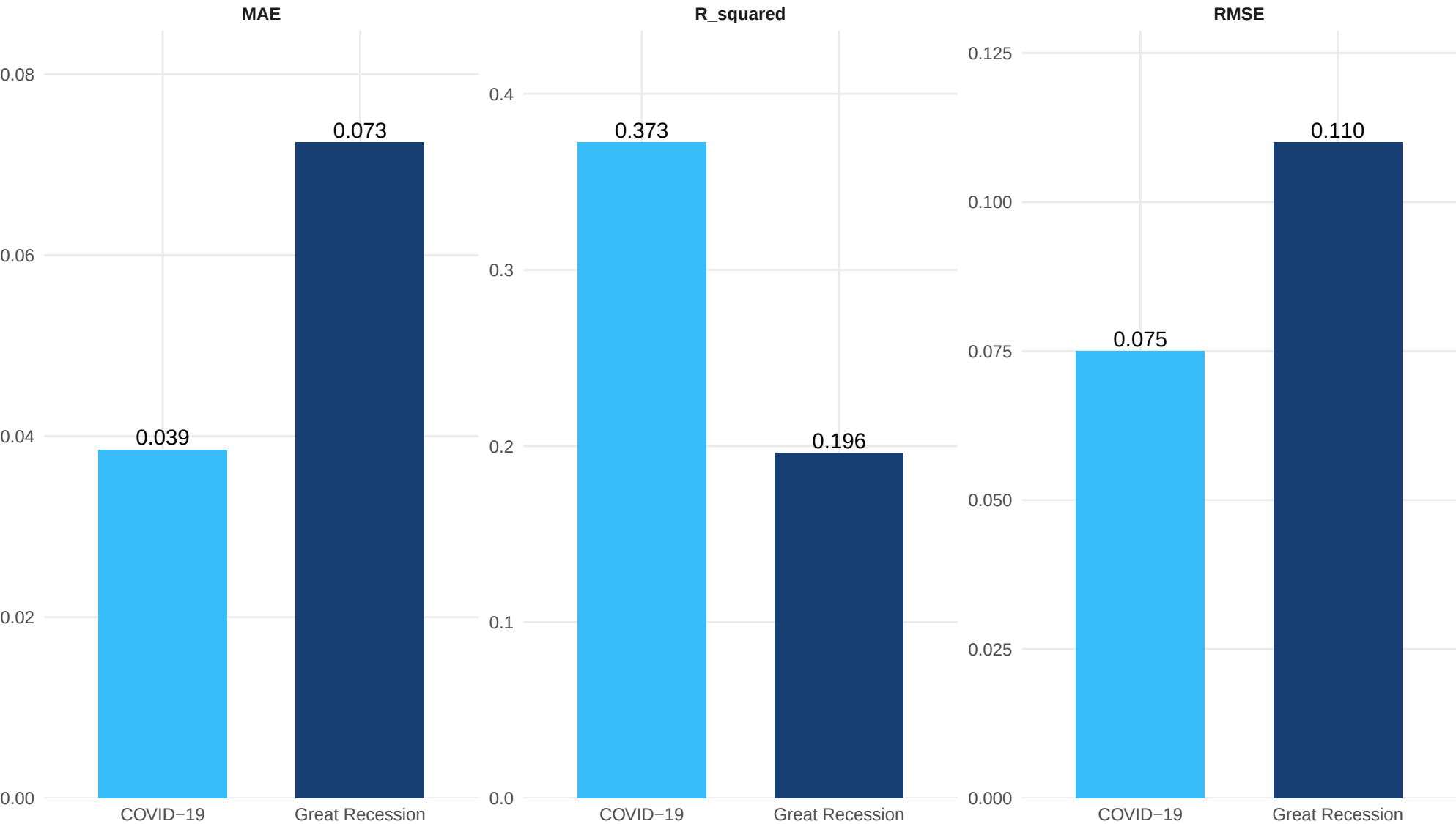
County convergence

OOB MSE stabilises as trees are added



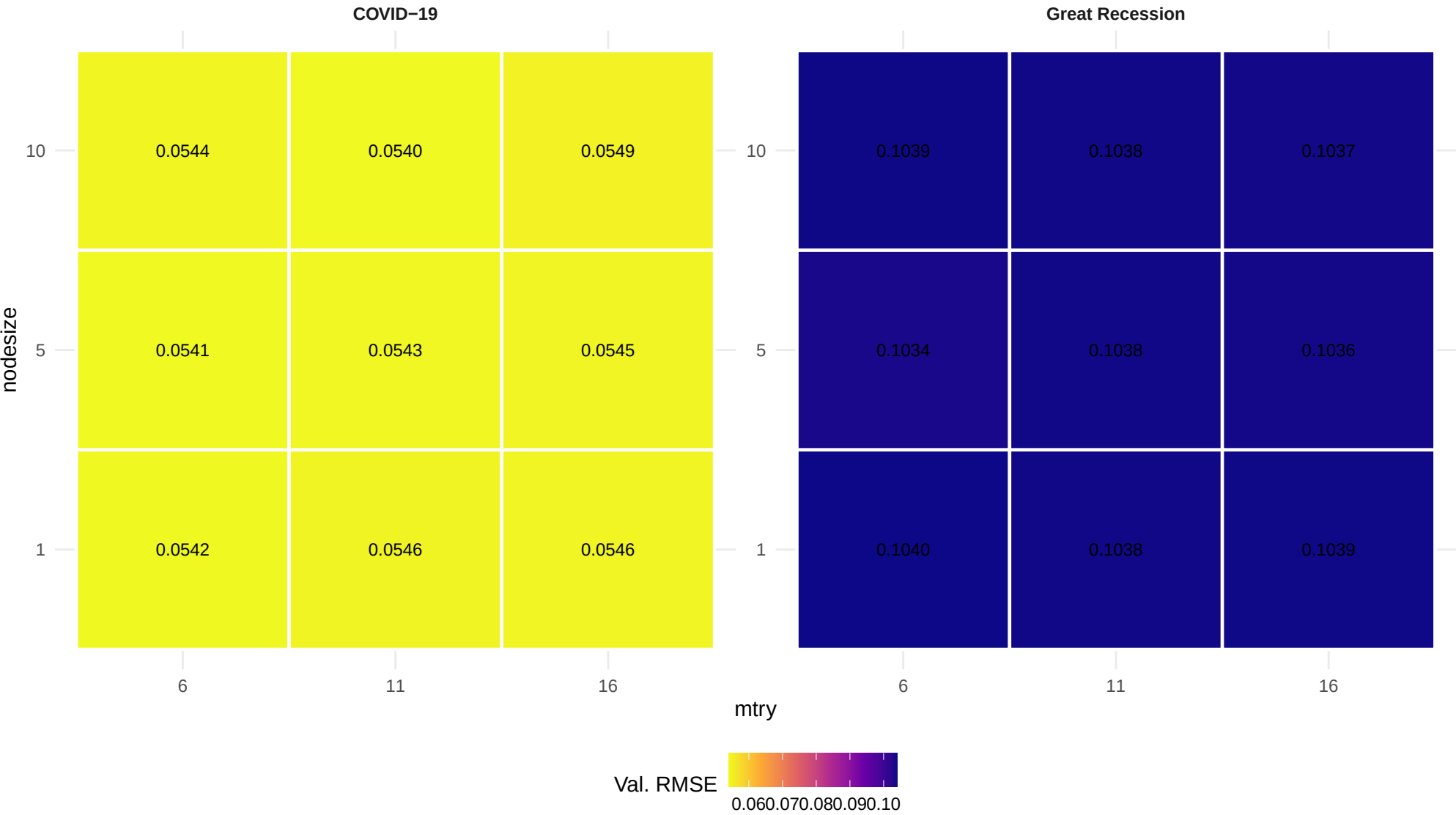
County performance

Lower RMSE/MAE better; higher R² better



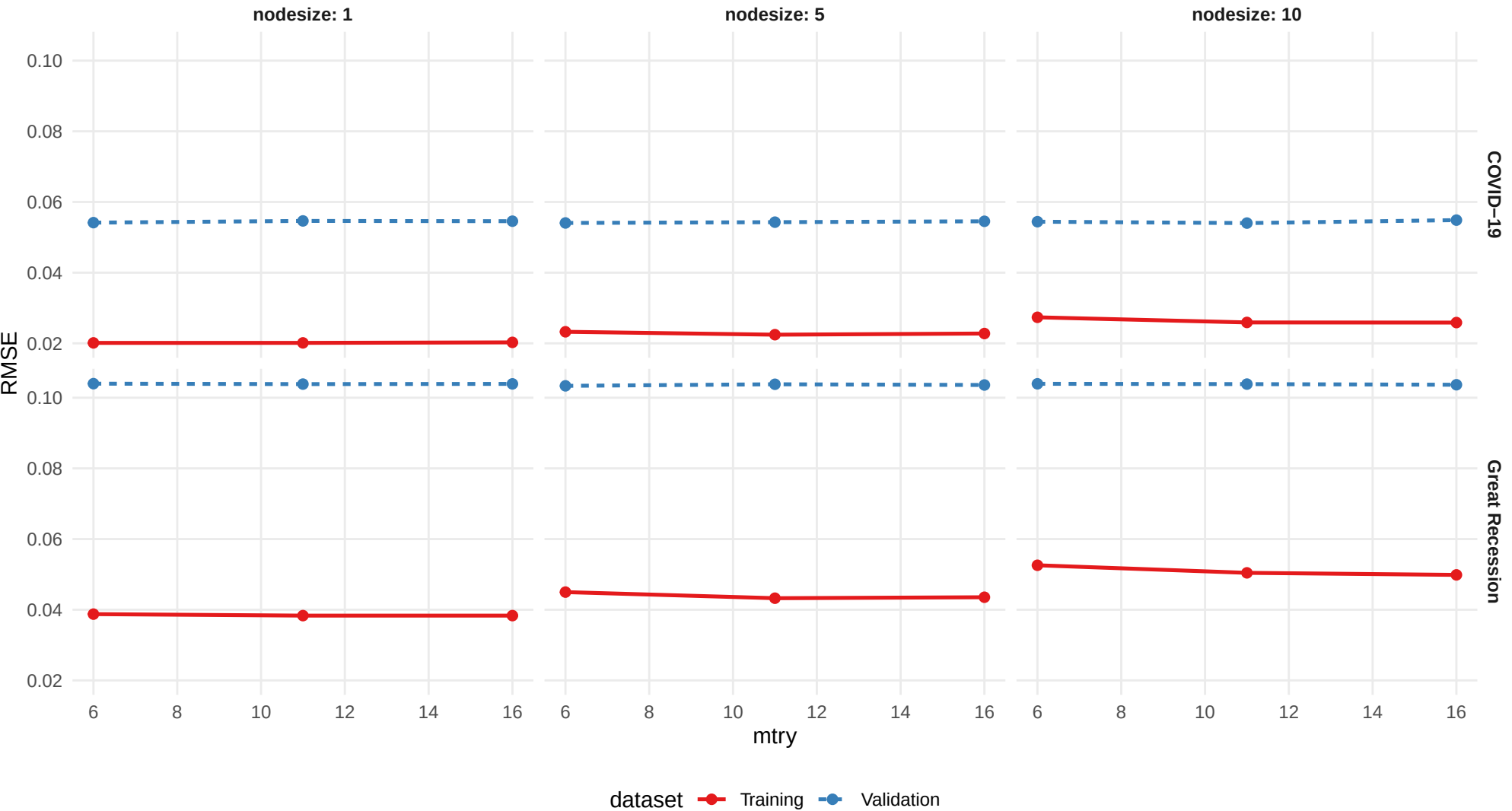
County hyperparameter tuning (val. RMSE)

Validation (OOB) RMSE per mtry / nodesize



County train/validation loss vs hyperparameters

Lines show RMSE for each nodesize across mtry



County Q-Q plot of residuals

Residual normality check

